Ans. Proportion $\frac{150}{250} = 60\%$ (part of the whole) ; change 250 $\frac{-200}{200} = 25\%$ (relative change).

4. Back · Table [IMG] Table of values of f :

x	0	1	2	3
$f(x)$	4	1	0	1

Image of 2 ? Preimage of 1 ?

Ans. $f(2) = 0$; preimages of 1: 1 and 3.

5. Running thread · MCQ [REP] On the globe, two points on the same parallel have:

A. the same latitude **B.** the same longitude **C.** the same distance at Greenwich **D.** the same altitude

Ans. A. the same latitude

Latitude: north/south angle to equator; longitude: east/west angle to Greenwich meridian.

Variations and extrema of a function

Sheet 43 Thu. 11 Feb. 2027 Increasing, decreasing, monotonic

1. Warm-up · Flash [IMG] $f(x) = -2x + 5$. Calculate $f(1)$ and $f(3)$; what do we notice?

Ans. $f(1) = 3$, $f(3) = -1$: the order of the images is the reverse of that of the inputs (f decreases).

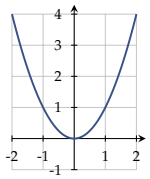
2. Core · MCQ [MOD] f is increasing on I means that, for all $a < b$ in I :

A. $a \leqslant b$ only **B.** $f(a) \geqslant f(b)$ **C.** $f(a) \leqslant f(b)$ **D.** $f(a) = f(b)$

Ans. C. $f(a) \leqslant f(b)$

Increasing: preserves order; decreasing: reverses it.

3. Core + · Graph [REG] Curve of $f(x) = x^2$ on $[-2;2]$:



Give the variation of f.

Ans. Decreasing on $[-2;0]$, increasing on $[0;2]$.

Expression used for the plot: f $(x) = x^2$.

4. Back · Flash [PRO] What multiplier corresponds to a 15% increase? To a decrease of 15 % ?

Ans. 1,15 ; 0,85.

5. Running thread · Flash [DIV] 457 divisible by 3? By 9? Justify.

Ans. Sum of digits $= 16$, not a multiple of 3: divisible neither by 3 nor by 9.

Sheet 44 Mon. 1 st March 2027 Table of variations ; maximum and minimum over an interval

1. Warm-up · Flash [IMG] Calculate x^2 for $x = -2, 0, 3$.

Ans. 4 ; 0 ; 9.

2. Core · Method [REG] f is decreasing on $[-3;1]$ with $f(-3) = 5$, $f(1) = -1$, then increasing on $[1;4]$ with $f(4) = 4$. Give the maximum and minimum of f on $[-3;4]$.

Ans. Maximum 5 reached at $x = -3$ $(5 > 4)$; minimum -1 attained at $x = 1$.

Compare the values at the endpoints of the interval AND at the changes of variation.

3. Core + · True/false [REG] True or false? Justify. The maximum of a function on $[a;b]$ is always reached at a or in b.

Ans. False. Counterexample: $f(x) = -x^2 + 4$ on $[-2;2]$: maximum 4 attained at 0.

4. Back · Table [PNP] Is this a table of proportional values?

x	1	2	3
y	5	9	13

Ans. No: $\frac{5}{1} \neq \frac{9}{2}$ (constant increase) $+4$ but $y(0) \neq 0$).

5. Running thread · Method [GEO] Is the triangle with sides 5, 12 and 13 right-angled? Which property should be used?

Ans. $5^2 + 12^2 = 169 = 13^2$: converse of Pythagoras' theorem, right-angled (right angle opposite 13).

Sheet 45 Tue. 2 March 2027 Affine functions: rate of change and variations

1. Warm-up · Flash [IMG] $f(x) = 3x - 1$. Calculate $\frac{f(4) - f(1)}{4 - 1}$.

Ans. $\frac{11 - 2}{3} = 3$: this is the slope m.

2. Core · Flash [MOD] Direction of variation of $f(x) = -4x + 7$?

Ans. $m = -4 < 0$: f is strictly decreasing on $\mathbb{R}$.

3. Core + · Proof [LIT] Show that $f(x) = 5x - 2$ is increasing on $\mathbb{R}$.

Ans. For $a < b$: $f(b) - f(a) = 5(b - a) > 0$, therefore $f(a) < f(b)$.

Model write-up for the proof required by the curriculum.

4. Back · Table [LEC] Ages of the members of a club.

Age	10	11	12	13
Count	3	6	10	6

How many pupils are at most 11 years old? What frequency?

Ans. $3 + 6 = 9$ pupils ; $\frac{9}{25} = 36\,\%$.

5. Running thread · Program → formula [REG] Procedure: multiply by 3 then subtract 2. Give $f(x)$ then $f(4)$.

Ans. $f(x) = 3x - 2$; $f(4) = 10$.

Sheet 46 Thu. 4 March 2027 Absolute value and square functions: sign and variations

1. Warm-up · Flash [CAL] Calculate $|-5|$, $(-5)^2$ and $|2 - 7|$.

Ans. 5 ; 25 ; 5.

2. Core · Method [REG] Give the variation table of $x \mapsto |x|$ and its sign.

Ans. Decreasing on $]-\infty; 0]$, increasing on $[0; +\infty[$; minimum 0 at 0 ; always positive.

3. Core + · Reflection [CMP] Compare $(-3)^2$ and $(-2)^2$ then 3^2 and 2^2. Is the square function increasing on $\mathbb{R}$?

Ans. $9 > 4$ in both cases, but $-3 < -2$: the square reverses the order on $]-\infty; 0]$; no, not increasing on $\mathbb{R}$.

4. Back · Method [PRB] Two fair dice are rolled. Probability that the sum is 7?

Ans. 6 favourable cases out of 36: $\frac{1}{6}$.

5. Running thread · Method [ALG] Procedure: choose a number; add 1; square it; subtract the square of the starting number. What result for 6? What general expression?

Ans. For 6: $49 - 36 = 13$. In general $(x + 1)^2 - x^2 = 2x + 1$.